
ShadowWeave: A Calibrated World Model of Occluded Space from a Single Depth Frame

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Abstract

1 An embodied agent must model the physical scene it cannot directly see: the
2 occupancy occluded by the obstacles it is moving among (the “shadow”), and
3 a world model of only the visible surface is blind to half of its workspace. We
4 ask how well a model can **complete** occupancy there from a **single monocular-**
5 **depth frame**, and whether it knows *when it is guessing*. ShadowWeave projects
6 one depth frame into an egocentric bird’s-eye-view (BEV) occupancy grid, marks
7 unobserved cells with an explicit shadow mask, and predicts occupancy and per-cell
8 uncertainty inside that mask. On a MuJoCo indoor benchmark, the model beats a
9 persistence baseline by a micro-averaged shadow-IoU gain of **+0.31 to +0.36 at**
10 **a 5 s horizon** (95% per-episode bootstrap CIs clear of zero at every horizon), far
11 exceeds an observation-blind dataset prior, and its uncertainty is **better calibrated**
12 **inside shadow (ECE 0.043–0.047, at 5 s) than in directly observed space (0.058–**
13 **0.086)**. Framing this as world modeling of the unobserved, a decomposition
14 separating *amodal completion* from *temporal forecasting* shows the gain is almost
15 entirely completion, and the pure-forecasting residual is small ($\leq +0.007$ at 5 s).
16 Randomizing room geometry per episode leaves the gain statistically unchanged on
17 never-seen rooms, which rules out memorization of a fixed shell. The completion
18 path runs at 7 ms (p95), and end-to-end with an off-the-shelf monocular-depth
19 network it is ~ 18 Hz on 2017-era hardware. It feeds an eyes-free spatial-audio
20 interface as a demonstration, and we release the methodology as a reusable protocol
21 for honest occupancy-completion claims.

1 Introduction

23 Any embodied agent acting in the physical world must reason about space it cannot directly observe:
24 the occupancy occluded by the obstacles it is moving among. A world model of only the visible
25 surface is blind to half of its workspace, and completing that hidden state is a prerequisite for
26 reasoning about what happens next. For any downstream use, that inference is trustworthy only
27 with an honest estimate of its own reliability, since a confident hallucination of clear space behind a
28 wall is worse than a calibrated “I don’t know.” Prior work addresses parts of this problem but not
29 their conjunction (§5). **ShadowWeave**, a world model of occluded space, produces from a single
30 monocular-depth frame (i) an egocentric BEV occupancy grid, (ii) an explicit shadow mask marking
31 cells the sensor cannot see, and (iii) a calibrated per-cell occupancy probability inside that mask. Our
32 contributions are:

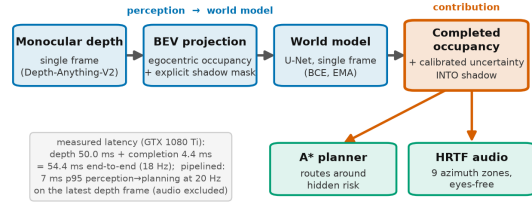


Figure 1: Pipeline: depth frame $\rightarrow$ 96×96 egocentric BEV occupancy and visibility; thresholded visibility gives the shadow mask; a 31M U-Net predicts occupancy at 1/3/5/10 s inside it, feeding A* planning and nine-zone HRTF audio.

- **A calibrated amodal-completion result.** From one depth frame, the model completes occluded occupancy with a micro-averaged shadow gain of $+0.31$ – 0.36 at 5 s (CIs clear of zero at every horizon), and its in-shadow uncertainty is *better* calibrated (ECE 0.043–0.047 at 5 s) than in observed space.
- **A decomposition methodology** that separates completion from forecasting. It reveals that the gain is spatial completion of persistent hidden structure; the pure-forecasting residual is at or near zero (fixed static $+0.000$; moving tier $+0.007$ [$+0.002$, $+0.012$] at 5 s, which is significant but practically small, and decays with horizon as real forecasting should). We report this negative honestly, since most pooled occupancy numbers in the literature share exactly this confound.
- **A memorization confound-kill.** Randomizing room geometry per episode leaves the gain statistically unchanged on never-seen rooms ($+0.306$ fixed vs $+0.309$ [$+0.294$, $+0.324$] randomized), showing completion is per-scene inference, not a memorized layout.
- **A real-time completion front-end** (7 ms p95 completion path, well inside 20 Hz; ~ 18 Hz end-to-end with off-the-shelf depth on 2017-era hardware), feeding an A* + nine-zone HRTF eyes-free interface as a systems demonstration (§4).

2 Method

A differentiable projector (Fig. 1) lifts each monocular-depth image (rendered in simulation; from a monocular-depth network in deployment) into an egocentric 96×96 BEV occupancy grid covering 5 m ahead, plus a **visibility** grid: cells along a ray up to the first returned surface are observed-free, the surface cell observed-occupied, and everything beyond is **shadow**. Thresholding visibility at 0.5 yields the shadow mask, the region of interest for every metric here; optical flow between BEV frames is an optional channel. A 31M-parameter U-Net predicts occupancy from the single-frame BEV stack at 1/3/5/10 s as per-cell probabilities, trained with positive-up-weighted binary cross-entropy (BEV targets are $\approx 7\%$ occupied) on 400 episodes with early stopping and EMA weights; we report the epoch-8 checkpoint (val. loss 0.300, IoU@5s 0.665). The input is a **single frame** with no multi-frame memory, which makes the completion-vs-forecasting distinction (§3) sharp. The per-cell probability *is* the uncertainty estimate; we test calibration **inside the shadow region specifically**, not merely grid-wide (where easy free space dominates). Downstream, the completed grid and its uncertainty drive an A* planner (extra cost through unobserved cells), a reactive collision-avoidance policy, and nine-zone HRTF audio cues; the audio interface is a demonstration, not user-evaluated.

3 Evaluation methodology

The empty-credit trap and micro-averaging. Our claim lives entirely inside a sparse, mostly-empty sub-region (shadow is $\sim 92\%$ never-occupied), and the natural masked IoU scores an empty prediction against an empty masked region as a perfect 1.0; since most shadow cells are legitimately empty, this inflates both model and baselines, non-uniformly across sub-masks. We therefore report **micro-averaged** shadow IoU: we pool integer intersection and union counts across all frames and

tier (fixed val, @5 s)	gain	95% CI
static	+0.306	[+0.288, +0.327]
moving	+0.363	[+0.349, +0.378]
debris	+0.315	[+0.281, +0.350]

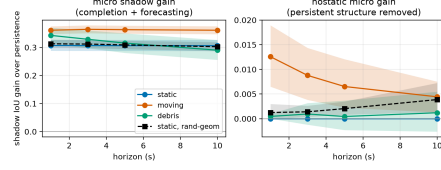


Figure 2: Micro shadow-IoU gain over persistence with 95% per-episode bootstrap CIs: per tier at 5 s, fixed geometry (left) and across horizons (right); every interval is clear of zero.

70 divide once (empty denominators are undefined, never 1.0), and the defended quantity is the **gain**
71 over a persistence baseline (the observed frame carried forward), not the absolute level.

72 **Completion vs. forecasting decomposition.** All horizons are rendered from the same issue pose
73 (static geometry projects to identical cells at every horizon), so each shadow cell is classified by its
74 occupancy pattern across horizons. **STATIC** cells are occupied at *all* horizons (persistent hidden
75 structure; $\approx 7\%$ of shadow), where recall measures **amodal completion**; **DYNAMIC** cells are
76 occupied at *some but not all* horizons ($\leq 1.5\%$), where IoU measures genuine **forecasting** beyond
77 persistence; and **NEVER** cells are occupied at *no* horizon ($\approx 92\%$), the false-positive test. (Dynamics
78 faster than the 1 s grid, for example debris settling within a second, fall into **STATIC**, biasing the
79 split toward completion; the continuously-moving tier is the cleaner forecasting test.) This separates
80 completion from clairvoyance, a distinction pooled shadow-IoU hides.

81 **Per-episode bootstrap CIs.** Frames within an episode share geometry and pose, so we resample
82 **episodes** with replacement ($B = 10,000$) and report the 2.5/97.5 percentile interval of the paired
83 model-minus-persistence pooled micro-gain difference across replicates for every headline gain.

84 **Calibration in shadow.** We measure expected calibration error (ECE) restricted to shadow cells,
85 contrasted with observed cells, using equal-mass binning (a fine 1000-bin histogram merged into
86 15 equal-count macro-bins, since in-shadow probabilities cluster near the prior and would leave
87 equal-width bins empty).

88 4 Experiments

89 **Setup.** A MuJoCo single-room benchmark with three tiers: **static** (fixed obstacles), **moving** (kine-
90 matic movers), and **debris** (objects that fall and settle). It has 400 training / 80 validation episodes on
91 disjoint seeds; per-tier analyses use 30-episode (9,000-frame) single-tier sets, randomized-geometry
92 validation uses 80 episodes, and fixed-geometry results use a 6×6 m room unless noted. Reproducibil-
93 ity is pinned: scene generation is bit-exact against a golden hash, and a rollout re-run reproduces
94 every accuracy metric bit-identically (hardware-dependent latency excepted).

95 **Shadow-gain headline.** The model beats persistence at every horizon (Fig. 2), with every bootstrap
96 interval clear of zero, so the gain is statistically significant, not sampling noise. In policy rollouts
97 the frame-averaged (macro) shadow gain is $+0.325/+0.322/+0.320/+0.178$ at 1/3/5/10 s. Raw macro
98 shadow-IoU levels (model 0.744 vs persistence 0.424 at 5 s) are empty-credit inflated and reported
99 only for reference; the micro-gain is the defended number.

100 **Observation-blind prior.** Persistence could be a weak baseline for *completion*, so we also evaluate a
101 pose-marginal dataset prior (mean training-set occupancy in the egocentric frame, no observation
102 at all), swept over thresholds and scored at its most favorable one. It reaches only 0.08–0.10 micro
103 shadow IoU across tiers (0.09 on the randomized-geometry set): to recall hidden structure (0.86–0.92)
104 it must blanket 72–79% of genuinely empty hidden space with false positives, whereas the model
105 achieves 0.61–0.72 IoU at a 1.4–3.4% false-positive rate. An observation-*conditioned* heuristic
106 (dilating observed obstacles into shadow, scored at its best radius) peaks at 0.31–0.35, at or below
107 persistence itself. Completion is thus learned observation-conditional inference; with geometry
108 randomization (below), this rules out memorized layouts, marginals, and local extrapolation alike.

Table 1: Top: shadow-mask decomposition at 5 s, fixed geometry (dynamic IoU spans 1 s→10 s): the gain is recall of persistent structure, not forecasting. Bottom: static-tier gain, fixed vs per-episode randomized geometry.

component (fixed val @5 s)	model	persistence
persistent-structure recall (amodal completion)	0.83–0.93	0.37–0.51
dynamic IoU, moving tier (1 s→10 s)	0.092 → 0.031	0.060 → 0.020
dynamic IoU, debris tier	0.020 → 0.019	0.006 → 0.023
false-positive rate on never-occupied cells	0.014–0.034	0.022–0.027

static tier, @5 s	micro shadow gain	95% CI	completion recall
fixed 6 × 6 geometry	+0.306	[+0.288, +0.327]	0.829
randomized geometry	+0.309	[+0.294, +0.324]	0.854

Completion, not clairvoyance. Decomposing the shadow mask attributes the gain almost entirely to amodal completion (Table 1). Excluding all persistent structure, the residual “nostatic” gain is at or near zero: fixed static +0.000, moving +0.007 (CI [+0.002, +0.012]), randomized-geometry static +0.002 ([+0.001, +0.004]) at 5 s. The honest claim is **amodal completion of hidden structure**, plus a small but statistically significant moving-tier forecasting signal that decays with horizon as genuine forecasting should. We do not claim dynamic forecasting on debris, where the dynamic signal is at the noise floor and persistence matches the model beyond 1 s.

Robustness to room geometry (memorization confound-kill). A fixed room shell could let the model memorize a constant wall prior, so we regenerated data with per-episode random width and depth (i.i.d. in [5, 8] m, obstacle counts scaled to keep ~7% positives), retrained from scratch, and evaluated on 80 held-out randomized rooms (Table 1, bottom): the gain does not drop (intervals overlap almost entirely) and completion recall is if anything higher on never-seen rooms, so completion is per-scene inference, not a memorized shell.

Calibration in shadow. Shadow is ~92% empty, so a raw in-shadow ECE is easy to make small; we therefore lead with the harder number. Excluding persistent structure entirely, in-shadow ECE at 5 s remains a low 0.055–0.056 on the tiers with dynamic content (the static tier’s structure-excluded region has no positive labels, hence no curve), so calibration is carried by the genuinely uncertain cells, not the easy always-occupied ones. On the full shadow mask at 5 s, in-shadow ECE is 0.043–0.047, lower than observed space (0.058–0.086), with Brier 0.022–0.036, though this edge is aided by shadow’s emptiness. Calibration is no fixed-room artifact: on randomized geometry, in-shadow ECE is 0.050–0.067 vs observed 0.056–0.077, and shadow is better-calibrated at 5–10 s (comparable at 1 s).

Ablations. Retraining without either auxiliary channel leaves the gain intact: no-visibility per-tier gains at 5 s (+0.309/+0.360/+0.313 vs full +0.306/+0.363/+0.315) are within CI and the no-flow rollout gain (+0.381 vs +0.320) is if anything higher; both are redundant (no optical flow on the critical path); and as independent retrainings that all land within the full model’s CIs, they also bound training-seed variance.

System. The perception-to-planning path runs at 6.47 ms p50 / 6.99 ms p95 (batch 1, deterministic U-Net), inside a 50 ms (20 Hz) budget; the excluded monocular-depth stage is not free (Depth-Anything-V2-Small: 49.9 ms p50 / 50.7 ms p95 in isolation, 54.4 ms measured end-to-end sequential, evaluation-era GTX 1080 Ti, 480×640), so deployment runs depth asynchronously while the 7 ms completion path consumes the latest depth frame at 20 Hz (audio synthesis excluded). An occupancy-arrival *diagnostic* (not falling-object anticipation; cell-level detection 0.787 at 2.62 s mean lead; component-level 0.752, precision only 0.714) counts events per due prediction, not per object, keys on cells revealed rather than newly occupied (occluded static geometry revealed by agent motion also counts), and its lead is bounded by the {1,3,5,10} s horizon grid, missing the ≥3 s target; we report it as a diagnostic, not a strength. Navigation reaches a per-episode collision rate of 0.367 (vs 0.60 untrained; straightness 0.773), missing the ≤0.10 target, so it is an honest demonstration and no more.

Generative variant. A conditional-diffusion (DDPM) world model is ongoing work: iterated DDIM sampling exceeds our batch-eval budget; a like-for-like comparison is future work.

5 Related work

Predicting the structure of unobserved space has been approached from three angles. Semantic scene completion, from SSCNet [Song et al., 2017] through MonoScene [Cao and de Charette, 2022] and its monocular successors VoxFormer [Li et al., 2023] and VisHall3D [Lu et al., 2025], infers dense voxel geometry, and even hallucinates geometry beyond the visible frontier, from a single image, but targets outdoor driving voxels evaluated purely by mIoU/IoU, without calibrated uncertainty in the occluded region. Occupancy world models such as OccWorld [Zheng et al., 2024], together with occupancy-forecasting benchmarks (Occ3D; Tian et al., 2023) and self-supervised raycasting forecasters [Khurana et al., 2023], instead predict the *temporal* evolution of multi-frame occupancy for autonomous vehicles. Closest in spirit, ProxMaP [Sharma et al., 2023] completes indoor proximal occupancy from a single view for navigation, yet emits only point estimates. ShadowWeave is distinct: egocentric BEV amodal completion into explicitly masked occluded space from one monocular-depth frame, with calibrated uncertainty inside the hidden region.

6 Conclusion and limitations

Calibrated amodal completion of occluded occupancy works from one monocular-depth frame, transfers to never-seen rooms, and fits a real-time perception-to-planning budget; the empty-credit-immune micro-gain, per-episode CIs, and completion/forecasting decomposition keep the claim honest and, we hope, get reused. The limitations are those it surfaces: evaluation is in simulation on clean simulator depth (robustness to estimated-depth noise is untested), so we make no quantitative real-world claim, though the front-end already accepts real monocular depth (Depth-Anything-V2), making sim-to-real transfer the natural next step; baselines are self-constructed (prior methods target different input regimes); the model performs amodal *completion*, not multi-step *forecasting*, and is framed accordingly; navigation misses its collision target and the audio interface is not user-evaluated (system context); and the occupancy-arrival lead is bounded by the horizon grid.

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